



University of Missouri - Columbia

AIAA Design, Build, Fly Proposal

2024 – 2025



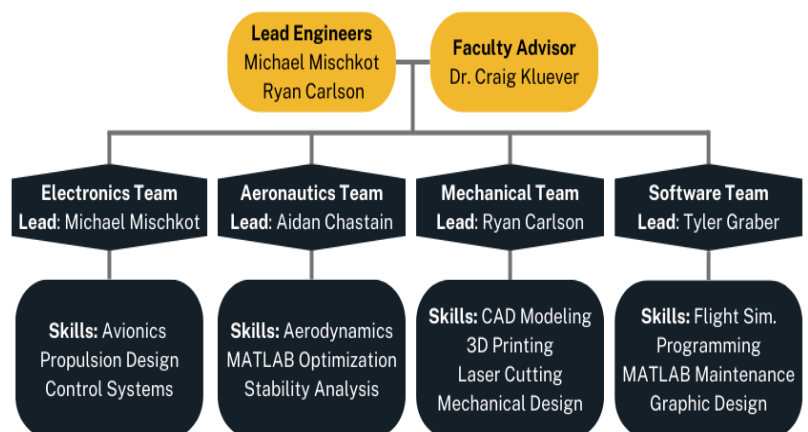
1. Executive Summary

This proposal details the approach of the **Mizzou AeroTigers**, representing the University of Missouri – Columbia, to design, analyze, manufacture, test, and fly a scale aircraft at the 2025 AIAA Design, Build, Fly Competition in Tucson, Arizona. This aircraft must be capable of executing missions centered around an *X-1 Supersonic Flight Program*, including a Delivery Flight, a Captive Carry Flight, and a Launch Flight, which requires the plane to release a glider that is capable of autonomous flight. The aircraft must also possess the ability to carry pylons and fuel tanks that are attached to the plane during the ground mission. After decomposing mission objectives and performing a sensitivity study on design parameters, an optimization program was developed to produce the highest-scoring aircraft. This year's design, **CAT** (Carrier for Autonomous Testing), will be a single-motor aircraft with conventional empennage and a high rectangular wing. It will be capable of carrying two pylons and fuel tanks, weighing a combined **4 lbs.** during Mission 2 (M2), and releasing an autonomous glider midflight during Mission 3 (M3). To execute this design, a project schedule, manufacturing plan, and testing plan have all been developed, based on the key milestones of conducting a maiden test flight in January, submitting a holistic design report in February, and attending the Fly-Off in April.

2. Management Summary

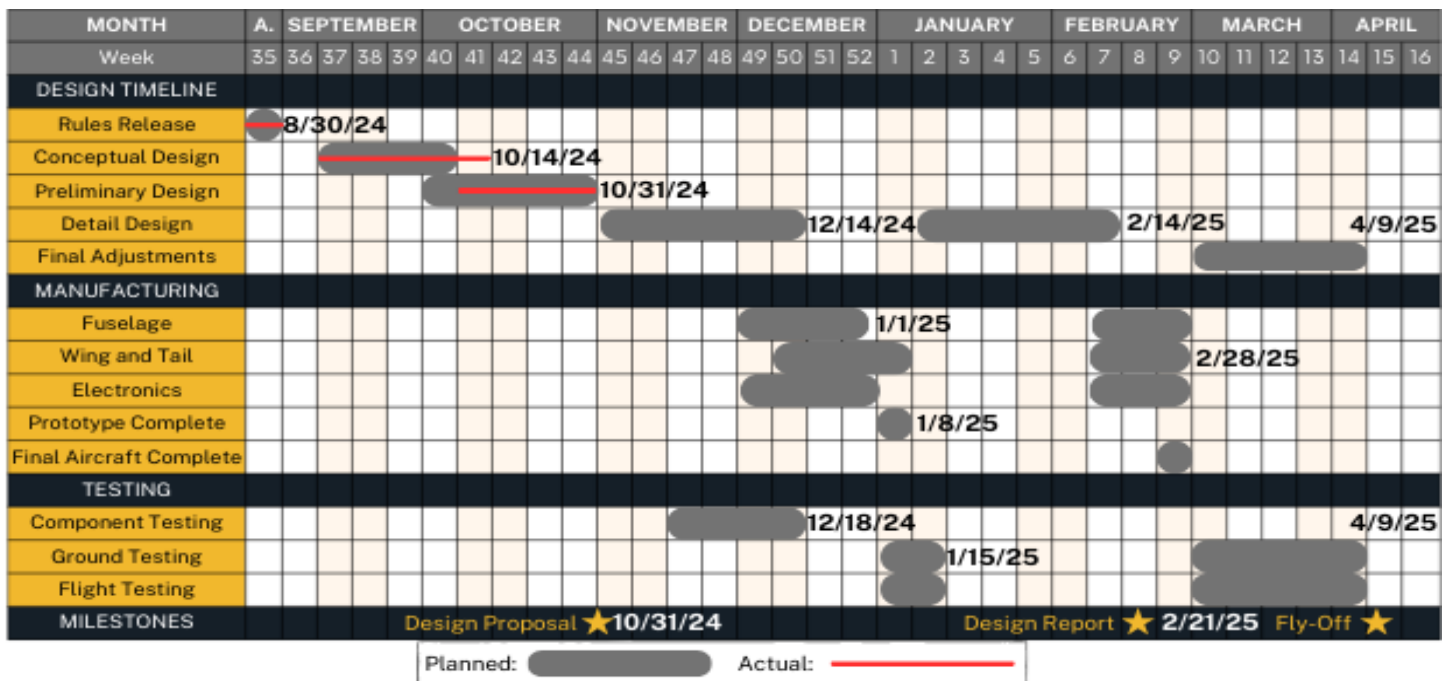
2.1 Team Organization: The Mizzou AeroTigers is an entirely student-led organization consisting of four teams: Electronics, Aeronautics, Mechanical, and Software. The teams are organized as shown in **Figure 2.1** and each has a student leader responsible for ensuring the team's timely success. The Electronics team is responsible for the design, testing, and validation of the propulsion, control, and energy storage systems. The Aeronautics team is responsible for the preliminary sizing, planform design, and aerodynamic analysis of the aircraft, which are critical to the optimization of mission deliverables. The Mechanical team is responsible for the design, SolidWorks modeling, and implementation of aircraft structures and mechanisms. The Software team is responsible for the setup and maintenance of the aircraft simulator, MATLAB optimization, and team webpage. All teams are involved in the manufacturing of their respective components, and in the writing of the proposal and the design report. Two lead engineers supervise and manage the project by creating timetables, integrating the work of the four teams, acquiring funding, recruiting new members, and leading general body meetings.

Figure 2.1: Team Organization



2.2 Schedule of Project Milestones: Figure 2.2 presents the AeroTigers' timeline to prepare for the Fly-Off. Maintaining scheduled deadlines is key to developing a high-scoring aircraft with ample time for prototyping and testing before the Fly-Off. The AeroTigers' conceptual and preliminary design phases are already complete; section three details these phases and the results. Currently, the detailed design phase is about to commence, involving the generation of comprehensive CAD models for manufacturing. Next, prototype manufacturing is scheduled to begin in early December, aiming for the first test flight to occur in mid-January. The data and experience gained from each flight will be used to improve the aircraft, with the final design scheduled to be fully manufactured by the end of February. The manufacturing strategy is further detailed in section four, while the testing procedures and planned testing schedule are contained in section five.

Figure 2.2: Competition Gantt Chart



2.3 Budget: The Mizzou AeroTigers' anticipated budget for the 2025 competition is detailed in **Figure 2.3**. The anticipated costs will be financed through a combination of fundraising, corporate outreach, and support from the University. This support will cover manufacturing, testing, and anticipated travel and shipping costs. This funding will enable the team to bring 10 students, including leadership and key contributors, to the in-person Fly-Off in Tucson, Arizona. The AeroTigers have already acquired the necessary manufacturing tools, including 3D printers, soldering equipment, and laser cutters.

Figure 2.3: Budget

Category	Item	Anticipated Cost (USD)
Electronics (\$1,920)	Propellers	\$250
	Transmitter, receivers	\$570
	Batteries	\$500
	Glider Autopilot With Redundancy	\$400
	Servos	\$200
Structures (\$1,300)	Carbon Fiber Spars	\$500
	MonoKote Film	\$200
	Landing Gear, Hardware	\$100
	Filament/Birch Plywood	\$250
	Fasteners/Glues/Tapes	\$100
	PLA Filament	\$150
Travel (\$6,550)	Lodging (5 nights)	\$1,400
	Airfare (Tucson, AZ)	\$4,250
	Aircraft Shipping	\$200
	Vehicle Rental	\$700
Total Anticipated Cost		\$9,770



3. Conceptual Design Approach

3.1 Mission Decomposition: This year's challenge is to develop an aircraft specializing in *X-1 Supersonic Flight*. The aircraft must comply with a **6 ft** total wingspan constraint and weigh less than **55 lbs**. The X-1 Test Vehicle must weigh less than **0.55 lbs.**, and all fuel tanks must have a minimum capacity of **16 fl. oz.** Three flight missions will be conducted to gauge the effectiveness of the aircraft design, and one ground mission will evaluate the efficiency of preparing the aircraft for flight from its unloaded configuration. **Figure 3.1** defines important design parameters and the scoring function for each mission, where M_n is the individual mission score, W_{fuel} is the payload weight, W_{veh} is the X-1 Test Vehicle weight, L is the number of laps, B is the amount of bonus points earned, and t is the mission completion time. To compute a total competition score, the individual mission scores will be summed and then multiplied by the total report score, which is derived from this formula: $TS = 0.15(P) + 0.85(R)$, where TS is the total report score, P represents the proposal score, and R represents the design report score. A small additional score for participation is also included. Flight order is determined by total report score and attaining a high flight priority is paramount to having enough time to complete all missions at the Fly-Off.

Figure 3.1: Mission Decomposition

MISSION	OBJECTIVE	SCORING	DESIGN PARAMETERS
M1: Delivery Flight	- Prepare for flight in under 5 minutes - Complete 3 laps in under 5 minutes	$M_1 = 1.0$	Balance CG in unweighted configuration
M2: Captive Carry Flight	- Complete 3 laps in minimum time - Carry heaviest total fuel weight - Carry X-1 Test Vehicle	$M_2 = 1.0 + \frac{[W_{fuel}/t]MU}{[W_{fuel}/t]_{Max}}$	- Maximize carrying weight - Maximize wing strength
M3: Launch Flight	- Complete maximum laps in 5 minutes - Release and land X-1 Test Vehicle in correct area - Carry potentially empty Fuel Tanks	$M_3 = 2.0 + \frac{[L + B/W_{veh}]MU}{[L + B/W_{veh}]_{Max}}$	- Maximize in-flight velocity - Balance X-1 Test Vehicle weight for optimal flight
GM: X-1 Flight Test Program Demonstration	Minimize time for attaching fuel tank pylons and fuel tanks.	$GM = \frac{t_{min}}{t_{MU}}$	Design pylons, fuel tanks, and X-1 Test Vehicle for efficient removal and attachment

3.2 Sensitivity Study: A sensitivity study was performed in MATLAB to analyze the impact of key scoring and design variables. **Figure 3.2** depicts how each mission objective impacts the scoring function. Two of the most critical criteria are lap speed for M2 and the total number of laps completed for M3, as both significantly influence the overall score. Takeoff weight, which affects the payload capacity in M2, and the weight of the X-1 Test Vehicle during M3 also play a smaller, but crucial role in scoring. The X-1 Test Vehicle weight would have a drastic effect on the bonus points earned, but because of the maximum weight constraint, the X-1 Test Vehicle weight only plays a minor role in the overall M3 scoring. GM time has an inverse, nonlinear relationship with score. **Figure 3.3** depicts how wing aspect ratio and takeoff weight impact the overall score at a constant thrust value. Larger aspect ratios reduce induced drag and allow for higher max speeds, meaning the necessary gross takeoff weight is lower for an equivalent score. By increasing gross takeoff weight, a lower aspect ratio can be utilized to increase structural stability and maneuverability. Overall, the mission score is more sensitive to increased aspect ratio than gross takeoff weight. The results from the sensitivity study were then used as inputs for generating the optimized preliminary design, with the overall process shown in **Figure 3.4**.

3.3 Preliminary Design: Based on the mission decomposition, an effective design must maximize lift, stability, and in-flight velocity while minimizing weight and configuration time. Thus, a monoplane with a high, rectangular wing was selected to optimize the M2 available lift, improve stability when under the loading conditions of the payload, and to minimize interference from the X-1 Test Vehicle. A tail-dragger landing gear configuration and conventional empennage were selected for strength, maneuverability, and ease of manufacturing. The NACA 4412 airfoil was selected out of available low-speed airfoils to maximize lift at the M2 velocity.

The preliminary design process was iterative and combined the scoring function generated from the sensitivity study with MATLAB's optimization toolbox to output the highest-scoring aircraft design available. The generated design parameters were then modeled in XFLR5 to output 3D lift and drag coefficients and stability metrics, which were then run back through the MATLAB optimization program until a converged solution was found. To maximize M2 and M3 efficiency, the program included a way to calculate the velocity for maximum L/D and the corresponding thrust required. From the analysis, key results included an aspect ratio of **7.0**, a takeoff weight of **18.0 lbs**, and a takeoff thrust of **21.8 lbs**. Based on the takeoff weight, a **4.0 lb** payload can be carried in M2.

The main requirement for the propulsion system for this year's aircraft is the capability of maximizing thrust to ensure the heaviest possible payload can be carried. Additional design considerations included cost, and the ability to reuse components from previous competition aircraft. Thus, the **Scorpion Power Systems A-5025-310kV** motor was selected. The motor will be placed in a tractor configuration. This motor is the same motor used in the 2023-2024 competition aircraft and is an **8S** brushless motor capable of spinning a **20 x 10E** propellor, and producing **24.4 lbs** of thrust. This motor is one of the most powerful available at the more affordable 8S level. The propulsion battery will be an 8S 3300 mAh LiPO. The electronic speed controller will be the **Spektrum Avian 120 Amp Brushless Smart ESC**.

Figure 3.2: Sensitivity of Scoring Variables

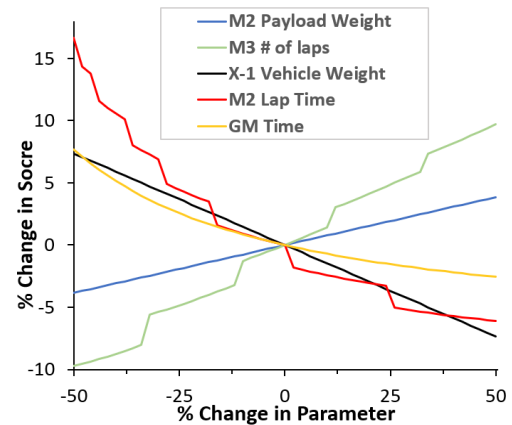


Figure 3.3: Analysis of Design Parameters

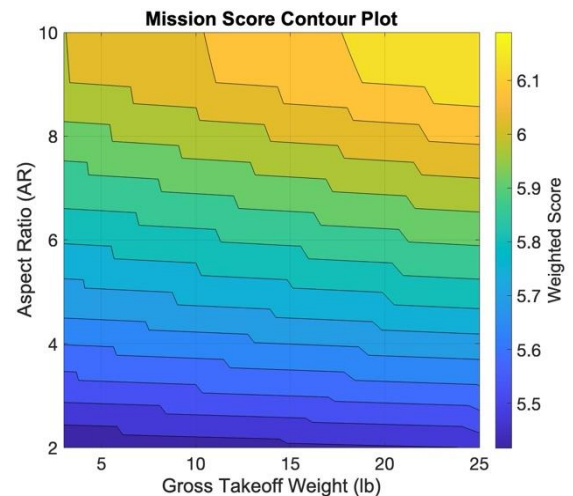
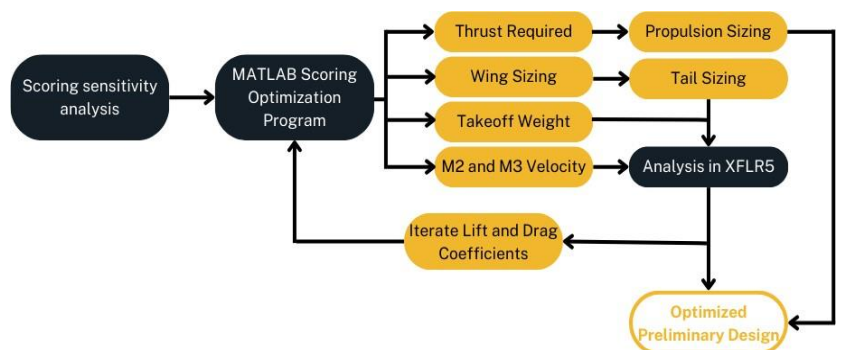
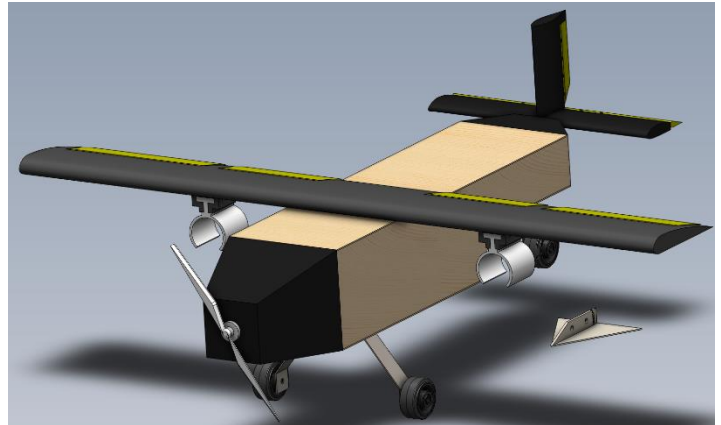


Figure 3.4: Preliminary Design Flowchart



A blended rectangular fuselage was selected to maintain a high fineness ratio while maintaining high manufacturability. Preliminary stability analysis, optimized for M2, resulted in an angle of incidence of 1.5° on the main wing and -2° on the tail, and the CG located at 25% MAC in the M2 configuration. A full conceptual model of the aircraft is seen in **Figure 3.5**.

Figure 3.5: Aircraft Conceptual Model



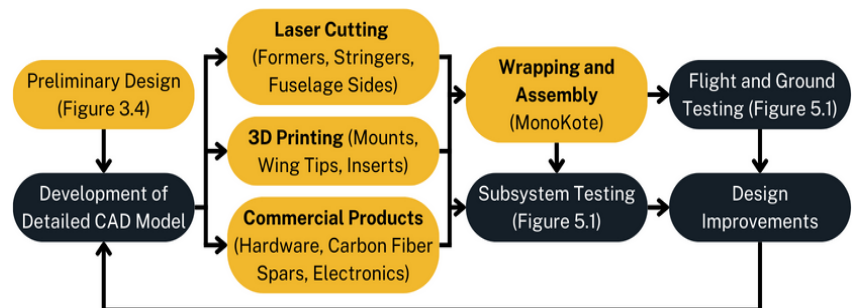
The initial design for the X-1 Test Vehicle, nicknamed Cub, is seen in the lower right of **Figure 3.5**. The X-1 will utilize a delta wing. The battery, autonomous flight system, and strobe light hardware will be affixed to the top of the wing. The X-1 will have two elevon control surfaces, controlled by linear servos. The X-1 will have a vertical mount extending from the middle of its fuselage that has two identical holes that will serve to attach the vehicle to the carrier plane. The X-1 will be attached to the carrier plane under the fuselage with two servo-actuated pins through the X-1's mounting holes. The X-1 will be made autonomous through the use of a flight controller equipped with ArduPilot software and flying wing model option. The GPS capability of the flight controller will be used to fly a descending spiral pattern over the bonus box. The strobing lights for the X-1 will be made with LEDs and an IC 555 timer in the astable oscillating mode with all necessary accompanying circuitry. The strobing light circuit will be powered by a low-capacity lithium battery. The power will remain disconnected from the X-1 until release through the use of an insulating ribbon anchored to the carrier plane. When the X-1 is released, this insulating ribbon will be pulled out of a contacting battery terminal on the X-1's circuitry that will then allow power to flow.

The initial design for the fuel tank pylons, shown underwing in **Figure 3.5**, centered on ensuring quick attachment and removal, securing the fuel tank during flight, and simplifying manufacturing and modification. Based on these parameters, early concepts included using either a buckle or a toothed rail system. For streamlined design and ease of use, a slide-and-bolt locking mechanism was ultimately selected. This allows the pylon to slide on and off with minimal effort while providing secure attachment from two mounting bolts. Additionally, the connection point between the wing and pylon is T-shaped, allowing for consistent bolt placement during installation.

4. Manufacturing Plan

The manufacturing flow is outlined in **Figure 4.1**. Weight, strength, and precision all must be considered throughout manufacturing. Due to the presence of the weighted fuel tanks on the wings, the strength of the wing-to-fuselage connection is critical. Each section of the aircraft will be broken down into individual components and modeled in

Figure 4.1: Manufacturing Flowchart



SolidWorks. The primary framework, including ribs, formers, and fuselage sides, will be cut from 1/8-inch plywood using a 60W CO2 laser cutter. More complex components, including pylons, pylon mounts, and wing tips, will be 3D-printed using PLA, with optimized infill patterns to enhance strength while minimizing weight. Key structural areas, such as the wing box and fuselage, will be reinforced with high-strength composite tubes to withstand flight loads. After manufacturing each subsystem, the aircraft will be assembled and sealed with a MonoKote exterior skin. The final competition aircraft will be designed and assembled in the same way with improvements made to the initial design from experience and testing.

5. Testing Plan

Before aircraft assembly, each team will subject their designs to thorough testing and validation. The propulsion system will be evaluated on a test stand to optimize thrust control and battery discharge rate. The electronic subassembly will be installed on a test bench to validate the wiring configuration. The pylon system and X-1 Test Vehicle supports will be tested to ensure stability, strength, and security during flight. Once the first full aircraft prototype assembly has been completed, each structure will be evaluated for strength and performance, with an emphasis on the technical inspection requirements. Ground testing will include a wing tip test, a fail-safe verification, a taxiing assessment, and a measurement of GM time. Flight testing will validate the X-1 Test Vehicle release for M3, assess battery endurance, and collect GPS data. The purpose, methodology, and due dates for each test are displayed in **Figure 5.1**. To maintain timely progress, the prototype aircraft will undergo its maiden flight in mid-January, followed by comprehensive testing throughout the remainder of January and February. In March, the final aircraft will be subjected to simulated competition runs, requiring the completion of a technical inspection, ground mission, and all three flight missions. Each flight date will allow the pilots to familiarize themselves with the aircraft, fine-tune the control systems, and optimize mission performance.

Figure 5.1: Testing Plan

CATEGORY	TEST	METHOD	GOAL	DUE DATE
Component Testing	Static Thrust Test	Measure thrust, battery usage, and power on a test stand	Select propeller/battery and create thrust model	12/04/24
	Electronics Bench Test	Attach all electronic systems on test bench	Ensure working avionics and propulsion subsystems	12/18/24
	X-1 Detachment Test	Test actuation of pinned X-1 detachment mechanism	Ensure reliability of X-1 detachment mechanism	12/18/24
	Pylon Strength Test	Attach maximally loaded fuel tank to pylon prototype and observe pylon reaction to loading	Confirm necessary strength of fuel tank pylons	12/18/24
Ground Testing	Landing Gear Test	Drop loaded aircraft onto landing gear	Ensure reliability of landing gear	1/08/25
	Wing Tip Test	Hold each mission-configured aircraft by its wingtips	Ensure structural stability and locate static margins	1/08/25
	Assembly Test	Test GM speed and consistency of installing fuel tanks and pylons	Select ground crew and optimize GM score	1/08/25
	Taxi Test	Test movement of aircraft on ground	Ensure landing gear functionality, balance, and steering capability	1/15/25
	Technical Inspection	Perform thorough inspections of the aircraft's construction	Demonstrate ability to pass technical inspection	1/15/25
Flight Testing	Takeoff Distance Test	Measure takeoff distance for different weights	Determine takeoff distance to assess possible hazards associated with long ground roll	1/15/25
	Flight Propulsion Test	Collect data in-flight to correlate thrust with velocity	Generate throttle curve and optimize flight velocity	1/15/25
	Flight Endurance Test	Measure battery usage in-flight with M3 configuration	Identify maximum laps possible before X-1 detachment	1/15/25
	Mock Flight Missions	Practice each flight mission	Ensure stability, optimize velocity, maximize payload	1/22/25